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5 NOV 1976
SUPERSEDING
MIL-C-20F
29 April 1975

MILITARY SPECIFICATION

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for established reliability (ER) and non-ER temperature-compensating, ceramic-dielectric, fixed capacitors for use primarily where compensation is necessary for circuit variations due to temperature changes, and in bypass and coupling applications. Capacitors meeting the ER requirements specified herein have a failure rate level ranging from 1.0 percent per 1000 hours to 0.001 percent per 1000 hours. These failure rate levels are established at a 90-percent confidence level based on the life test parameters specified and are maintained at a 10-percent producer's risk. An acceleration factor of 8:1 has been used to relate the life test data obtained at 200 percent of rated voltage at the applicable high test temperature to the rated voltage at the applicable high test temperature.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form, and as specified (see 3.1 and 6.2):

ER

CCR75	CH	1RO	C	M
Style (1.2.1.1)	Characteristic (1.2.1.2)	Capacitance (1.2.1.3)	Capacitance tolerance (1.2.1.4)	Failure rate level (1.2.1.5)

NON-ER

CC75	CH	1RO	C
Style (1.2.1.1)	Characteristic (1.2.1.2)	Capacitance (1.2.1.3)	Capacitance tolerance (1.2.1.4)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Aeronautical Systems Division, ASD/ENYES, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.1 Style. The style is identified by either the three-letter symbol "CCR" (ER parts) or the two-letter symbol "CC" (non-ER parts) followed by a two-digit number. The letters identify temperature-compensating, ceramic-dielectric, fixed capacitors, and the number identifies the shape and dimensions of the capacitor.

1.2.1.2 Characteristic. The characteristic is identified by a two-letter symbol in accordance with table I. The first letter identifies the nominal temperature coefficient; the second letter identifies the approximate tolerance envelope for the temperature coefficient (F = ±15 ppm/°C; G = ±30 ppm/°C; H = ±60 ppm/°C; J = ±120 ppm/°C; and K = ±250 ppm/°C). (See 6.5 and figures 3 to 11 inclusive.)

TABLE I. Characteristic.

Symbol	Nominal temperature coefficient	Symbol	Nominal temperature coefficient
	ppm/°C		ppm/°C
AH	100 ±60	PH	-150 ±60
AJ	100 ±120	PJ	-150 ±120
AK	100 ±250	PK	-150 ±250
CF	0 ±15	RF	-220 ±15
CG	0 ±30	RG	-220 ±30
CH	0 ±60	RH	-220 ±60
CJ	0 ±120	RJ	-220 ±120
CK	0 ±250	RK	-220 ±250
HF	-30 ±15	SG	-330 ±30
HG	-30 ±30	SH	-330 ±60
HH	-30 ±60	SJ	-330 ±120
HJ	-30 ±120	SK	-330 ±250
HK	-30 ±250	TG	-470 ±30
LF	-80 ±15	TH	-470 ±60
LG	-80 ±30	TJ	-470 ±120
LH	-80 ±60	TK	-470 ±250
LJ	-80 ±120	UG	-750 ±30
LK	-80 ±250	UH	-750 ±60
PF	-150 ±15	UJ	-750 ±120
PG	-150 ±30	UK	-750 ±150

1.2.1.3 Capacitance. The nominal capacitance value expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. When the nominal value is less than 10 pF, the letter "R" shall be used to indicate the decimal point and the succeeding digit(s) of the group shall represent significant figure(s). For example 1R0 indicates 1.0 pF; R75 indicates 0.75 pF; and 0R5 indicates 0.5 pF.

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1.2.1.4 Capacitance tolerance. The capacitance tolerance is identified by a single letter in accordance with table II.

TABLE II. Capacitance tolerance.

Symbol	Capacitance tolerance (+)
B	0.1 pF
C	0.25 pF
D	0.5 pF
F	1 percent
G	2 percent
J	5 percent
K	10 percent

1.2.1.5 Failure rate level. The failure rate level shall be as specified in table III.

TABLE III. FR level (established at a 90-percent confidence level).

Symbol	FR level Percent/1,000 hr
M	1.0
P	0.1
R	0.01
S	0.001

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-39028 - Capacitors, Packaging of.
MIL-C-45662 - Calibration System Requirements.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Reliability (ER styles only). Reliability of capacitors furnished under this specification shall be established and maintained in accordance with the procedures and requirements specified in MIL-STD-790 and MIL-STD-690 with details specified in 4.1.1, 4.4.4, and 4.4.6.

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Insulating and impregnating compounds. Insulating and impregnating compounds, including varnishes, waxes, resins, and the like, shall be suitable for each particular application. Compounds shall preserve the electrical characteristics of the insulation to which they are applied.

3.5 Design and construction. Capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 Case. Capacitors shall be effectively sealed against the entry of moisture. When a molded case is specified (see 3.1), the capacitor element shall be enclosed via transfer molding or the use of a pre-formed case. When a conformal (dipped) case is specified (see 3.1), the capacitor element shall be enclosed with insulating resin, plastic, or ceramic.

3.5.2 Connections. Electrical connections shall not depend upon wires, lugs, terminals, and the like, which are clamped between a metallic member and an insulating material other than the ceramic material. Such connections shall be soldered or shall be clamped between metallic members.

3.6 Thermal shock and voltage conditioning (ER parts only). When tested as specified in 4.7.2, capacitors shall withstand the extremes of high and low temperature without visible damage and meet the following requirements:

Insulation resistance:	
(At 125°C) - - - - - - - - - - - - -	Shall be not less than the value shown on figure 2.
(At 25°C) - - - - - - - - - - - - - -	Shall be not less than the value shown on figure 2.
Dissipation factor (at 25°C) - - - -	Shall not exceed the initial requirement (see 3.9).
Capacitance (at 25°C) - - - - - - - -	Shall be within the tolerance specified (see 3.1).

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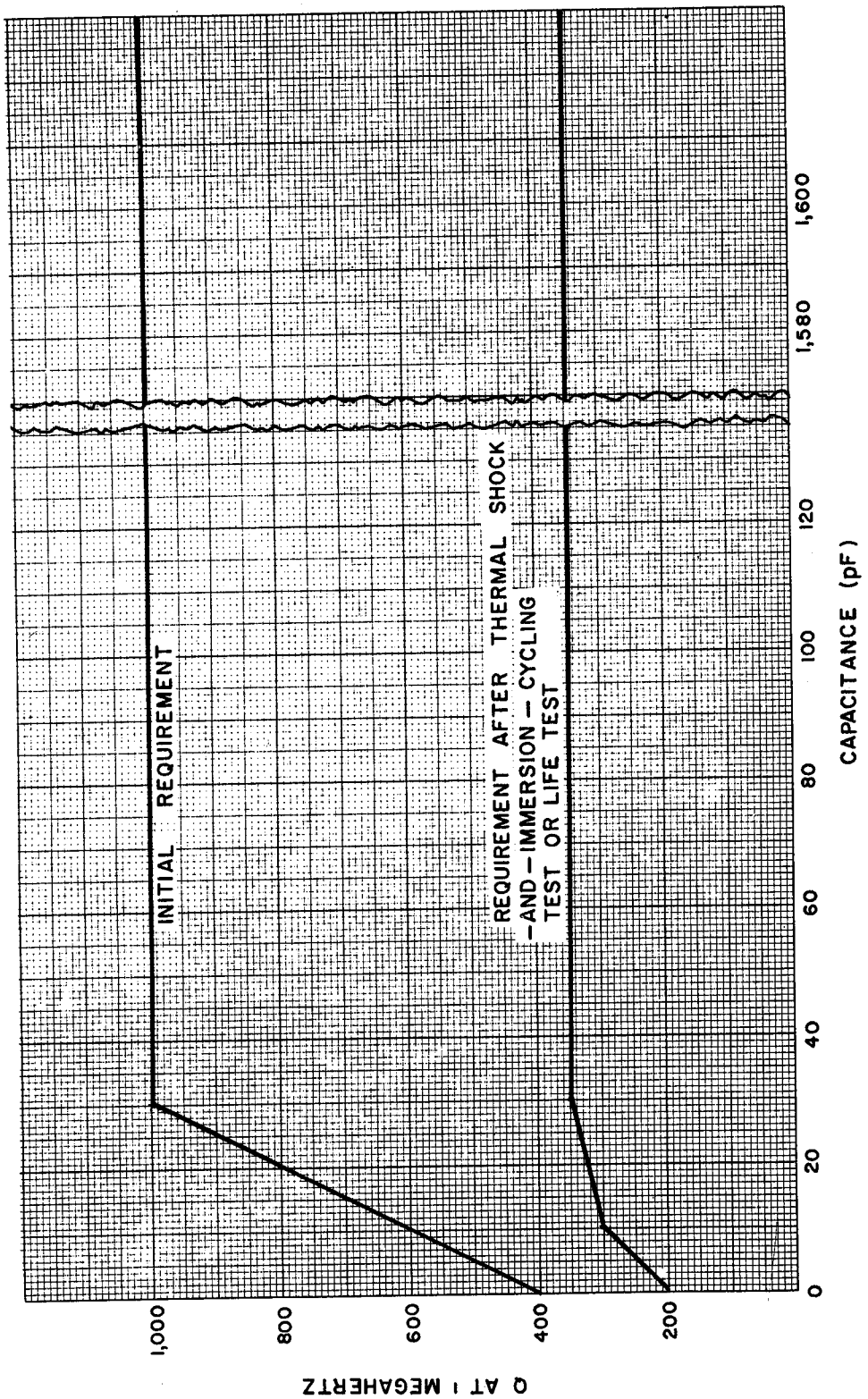


FIGURE 1. Minimum Q values.

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3.7 RF current (when specified, see 3.1). When determined as specified in 4.7.3, the capacitor shall not exceed the specified temperature with the rated RF current applied (see 3.1).

3.8 Capacitance. When measured as specified in 4.7.4, the capacitance shall be within the tolerance shown in the type designation (see 3.1).

3.9 Dissipation factor (when specified, see 3.1). When determined as specified in 4.7.5, the dissipation factor shall not exceed 0.15 percent.

3.10 Quality factor (Q) (when specified, see 3.1). When determined as specified in 4.7.6, the Q shall be not less than the value shown on figure 1.

NOTE: Due to limitations of measuring equipment, capacitors of high capacitance and high Q may appear to have negative or infinite values of Q, and zero or negative dissipation factors. These capacitors shall be considered as having met the Q requirements, provided they meet the requirements for dielectric withstanding voltage and insulation resistance as specified in 3.11 and 3.13, respectively. (See 6.4)

3.11 Dielectric withstanding voltage. When tested as specified in 4.7.7, capacitors shall withstand the dc potential without damage or breakdown.

3.12 Barometric pressure. When tested as specified in 4.7.8, capacitors shall withstand the dc potential without flashover.

3.13 Insulation resistance. When measured as specified in 4.7.9, the insulation resistance shall be not less than the value specified on figure 2.

3.14 Temperature coefficient and capacitance drift (see 4.7.10).

3.14.1 Temperature coefficient. Unless otherwise specified (see 3.1), each capacitance-temperature curve shall lie within the applicable tolerance envelope for the applicable temperature coefficient, as shown on figures 3 to 11 inclusive. (See 6.5.)

3.14.2 Capacitance drift. The capacitance drift shall be within ± (0.2 percent or 0.05 pF), whichever is greater.

3.15 Shock (specified pulse). When capacitors are tested as specified in 4.7.11, there shall be no momentary or intermittent contacts of 0.5 millisecond (ms) or greater duration, open- or short-circuiting, or other evidence of mechanical damage.

3.16 Vibration, high frequency. When tested as specified in 4.7.12, there shall be no momentary or intermittent contacts of 0.5 ms or greater duration, open- or short-circuiting, or other evidence of mechanical damage; and capacitors shall meet the following requirements:

Capacitance - - - - - - - - - -	Shall change not more than ± (2 percent or 0.5 pF) (0.25 pF for values less than 10 pF), whichever is greater, from the initial value obtained when measured as specified in 4.7.4.
Dissipation factor or Q (as applicable, see 3.1) - - -	Shall not exceed the initial requirement.
Dielectric withstanding voltage	Shall be as specified in 3.11.
Insulation resistance - - - - -	Shall be as specified in 3.13.

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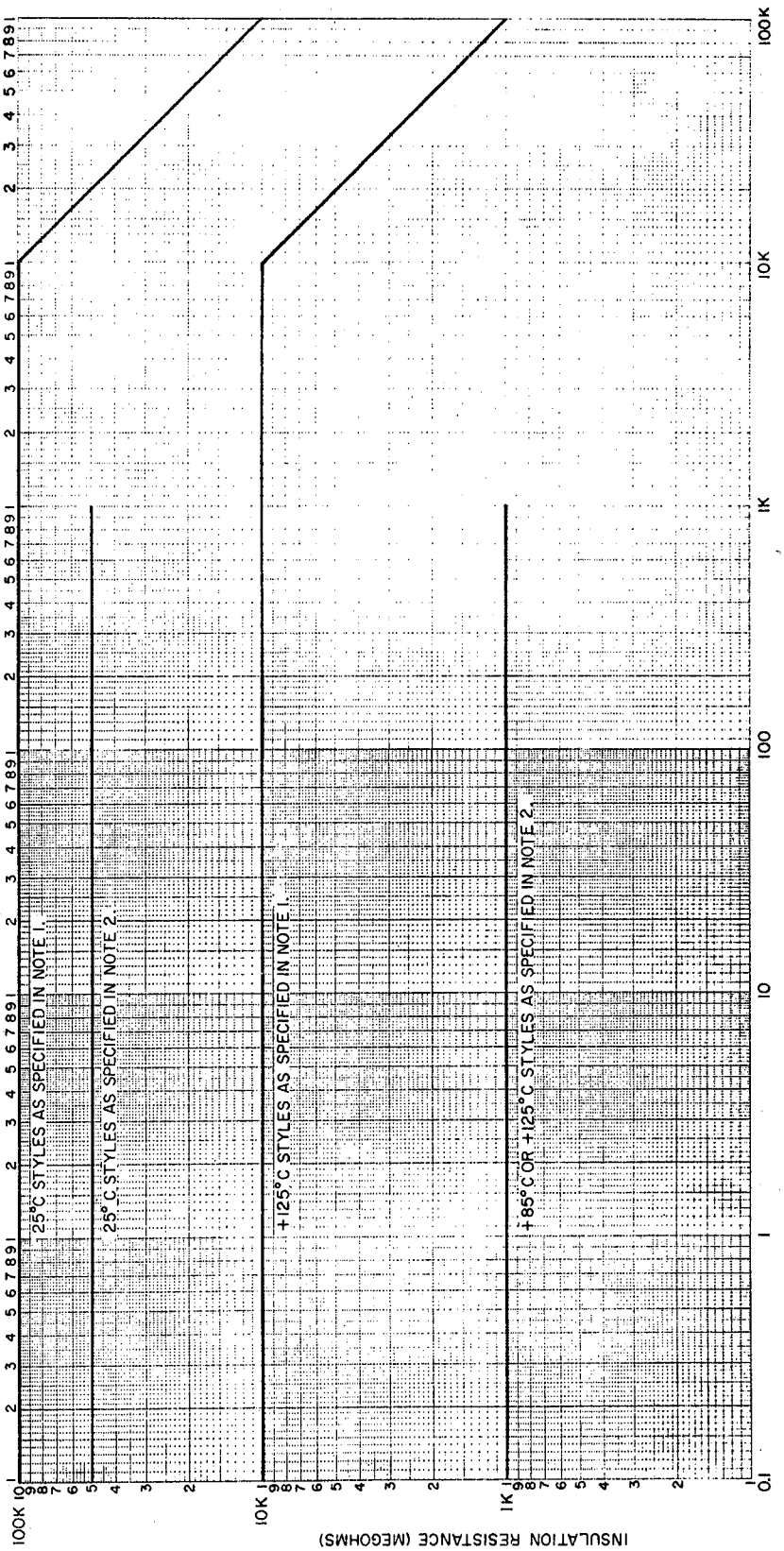


FIGURE 2. Insulation resistance versus capacitance.

NOTES:
1. Applicable to ER styles CCR05, 06, 07, 08, 13, 14, 15, 16, 17, 18, 54, 55, 56, 57, 75, 76, 77, 78, 79, 81, 82, and 83 and non-ER styles with same two-digit number.
2. Applicable to non-ER styles CC20, 21, 22, 25, 26, 27, 30, 31, 32, 33, 35, 36, 37, 45, 47, 50, 51, 52, and 53 only.

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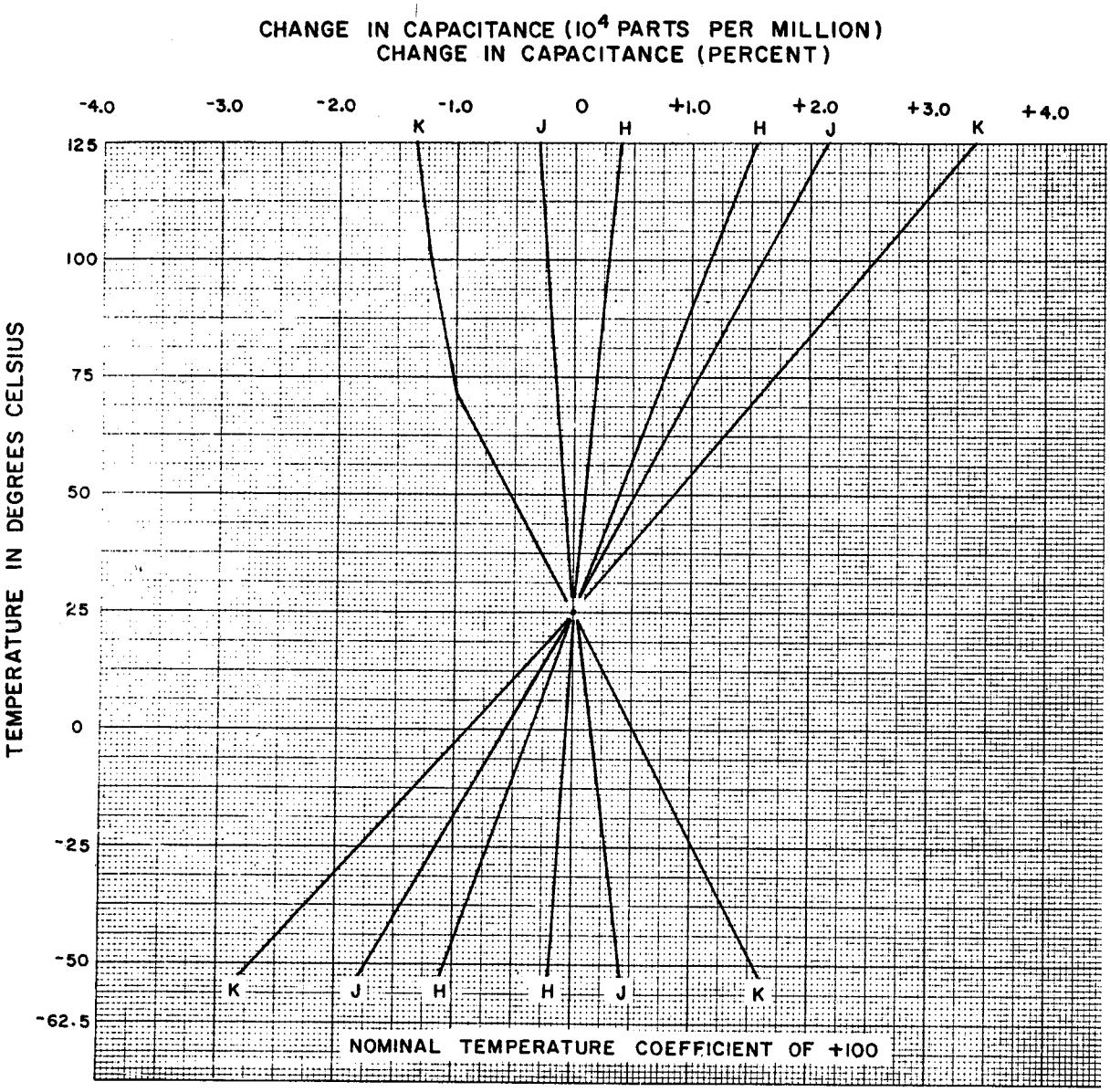


FIGURE 3. Tolerance envelopes for temperature coefficient A.

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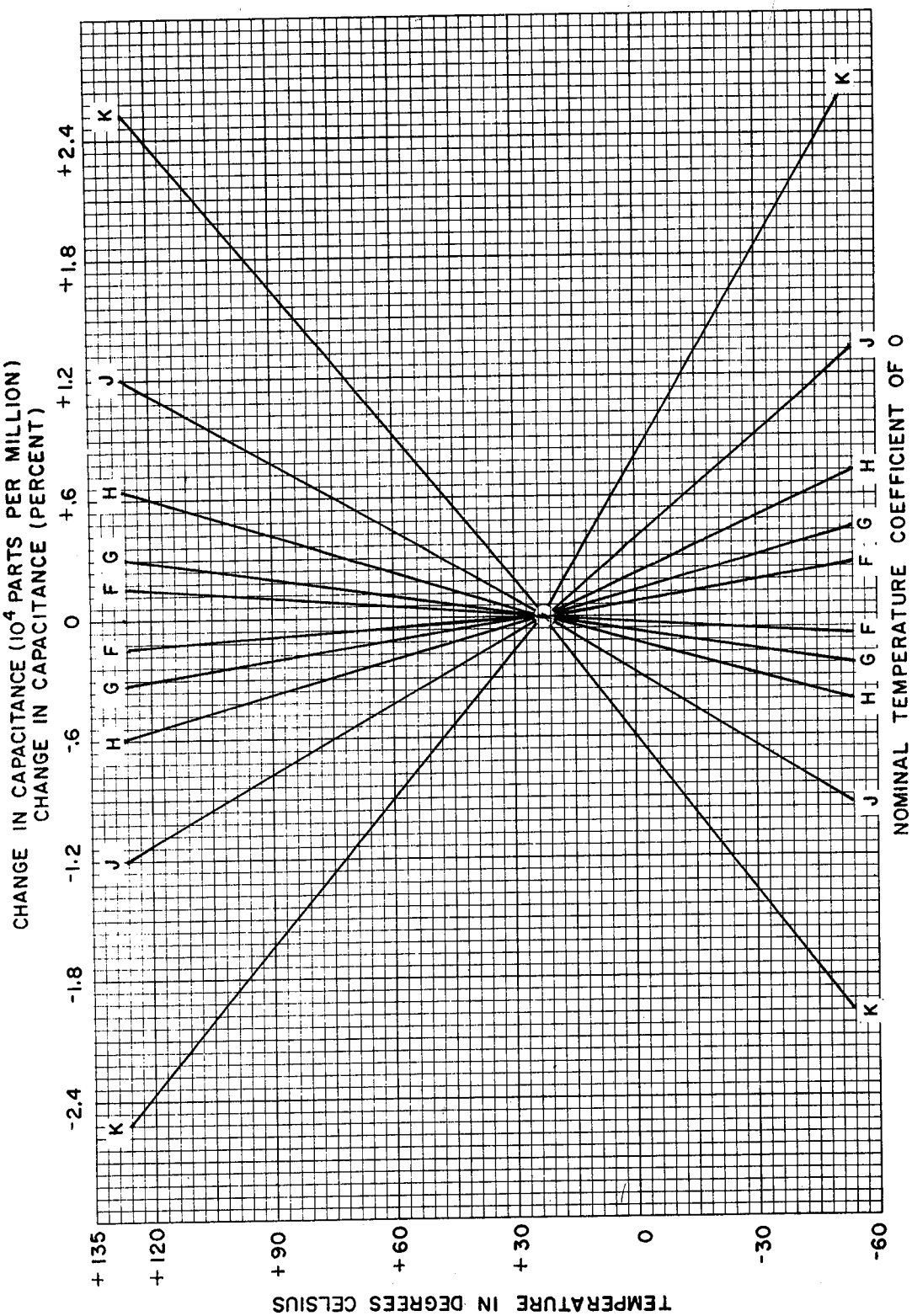


FIGURE 4. Tolerance envelopes for temperature coefficient C.

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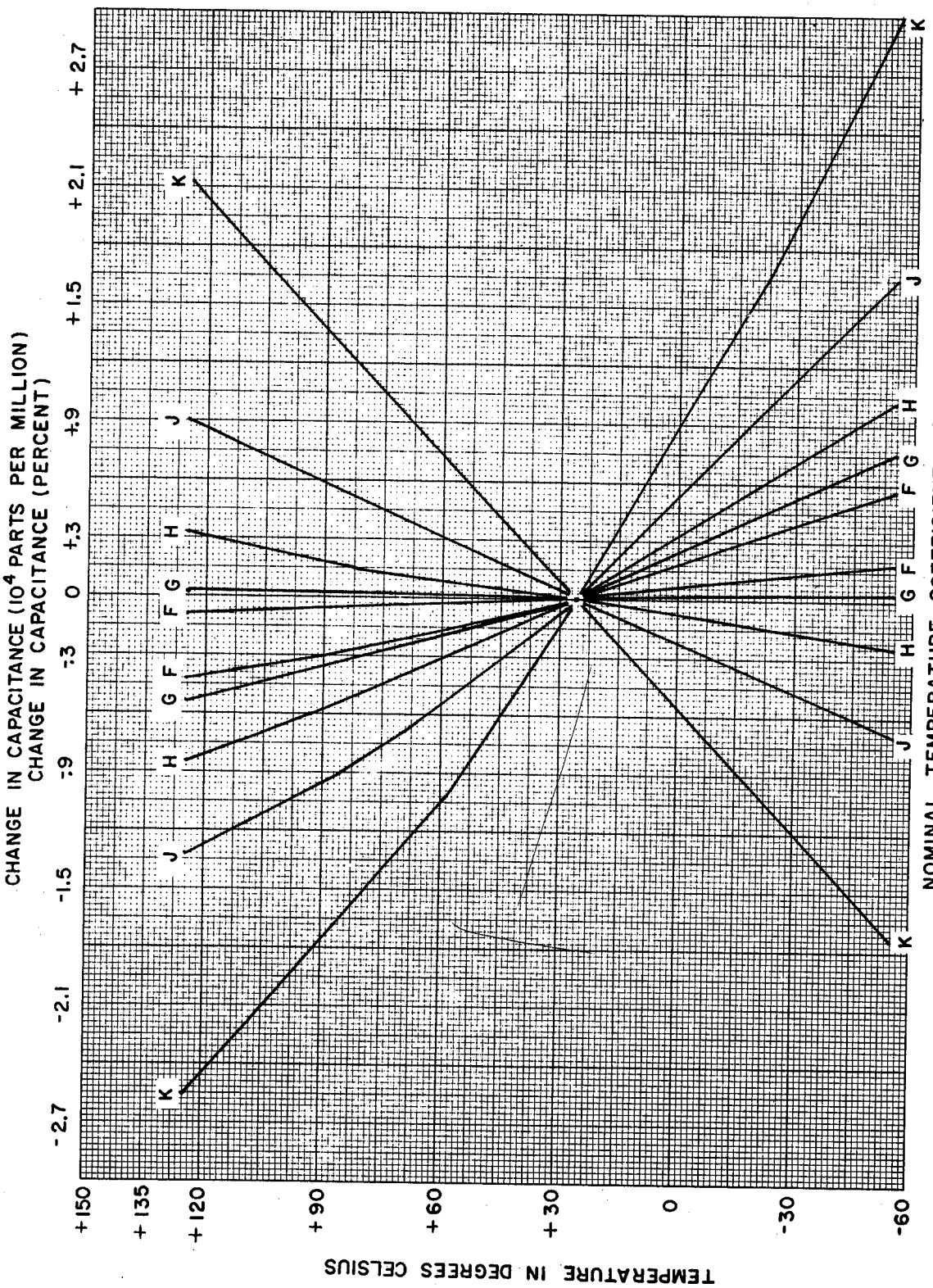


FIGURE 5. Tolerance envelopes for temperature coefficient H.

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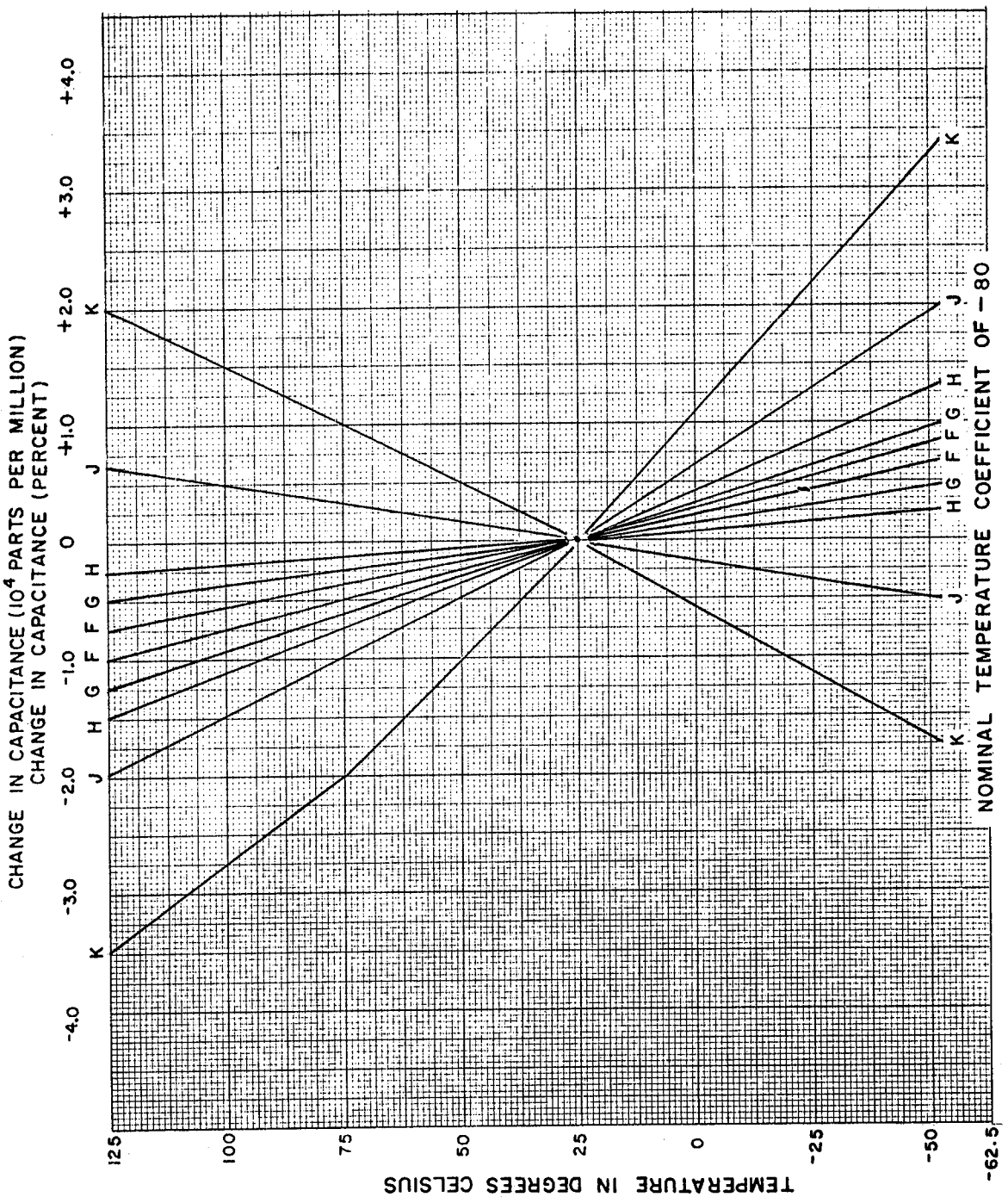


FIGURE 6. Tolerance envelopes for temperature coefficient L.

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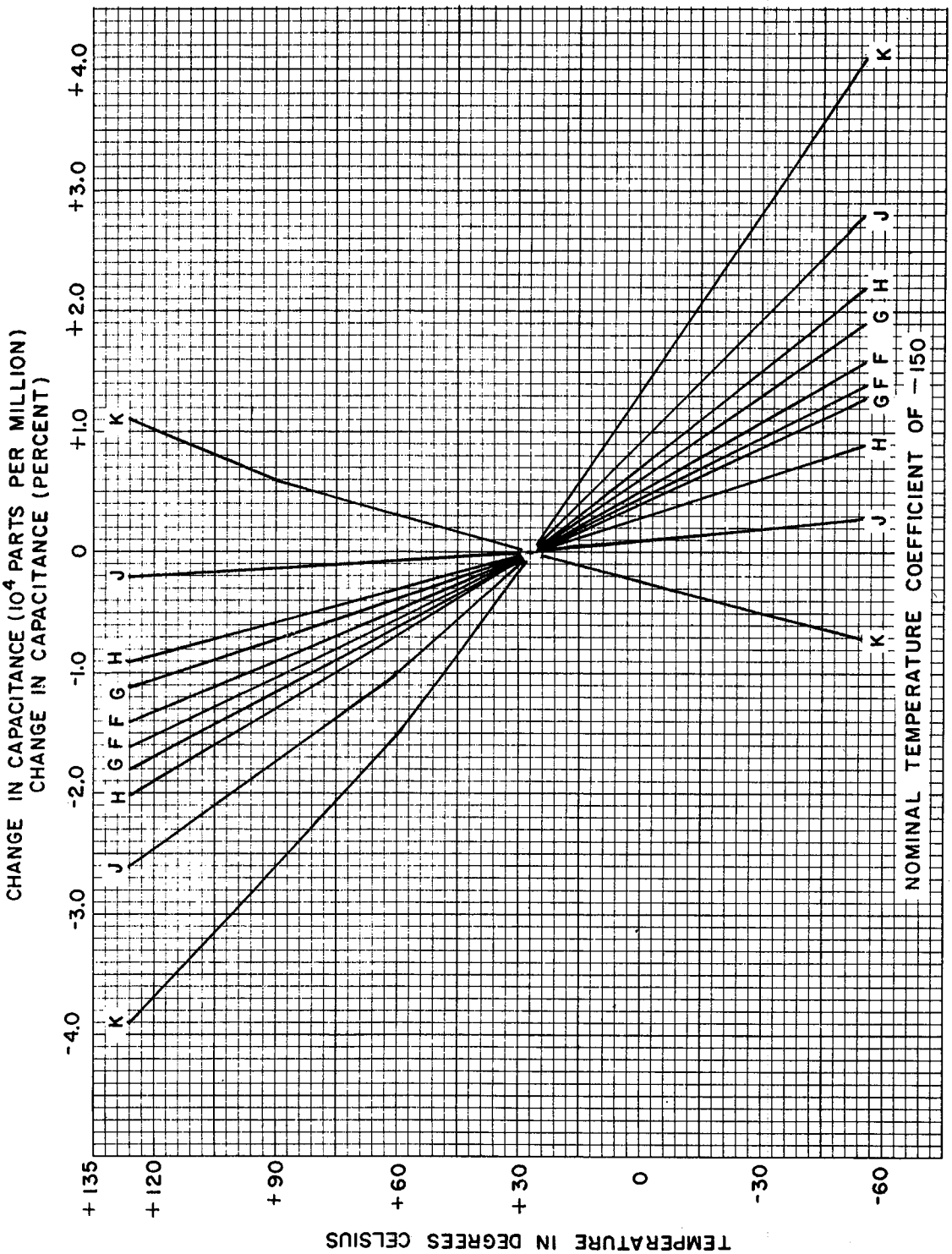


FIGURE 7. Tolerance envelopes for temperature coefficient P.

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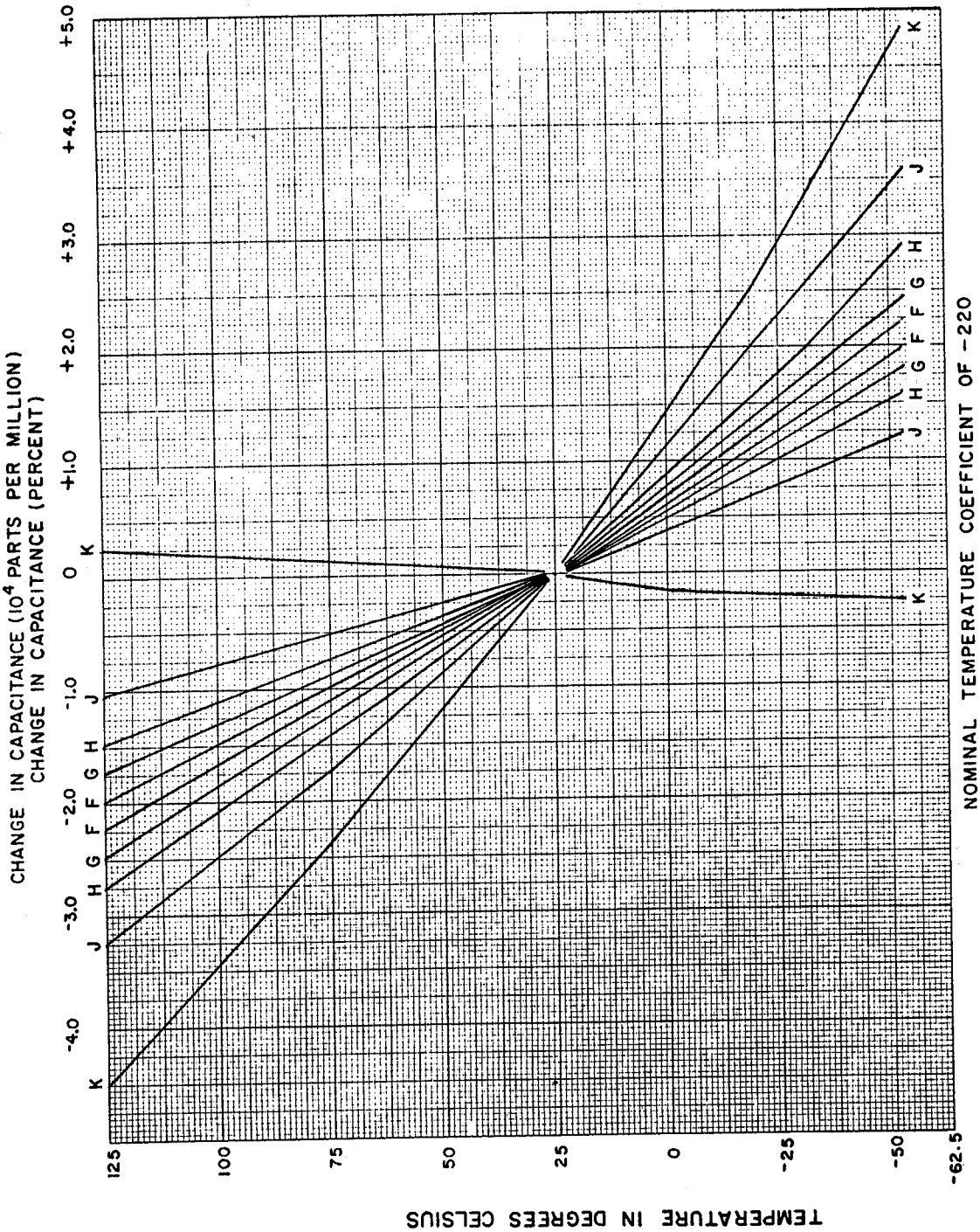


FIGURE 8. Tolerance envelopes for temperature coefficient R.

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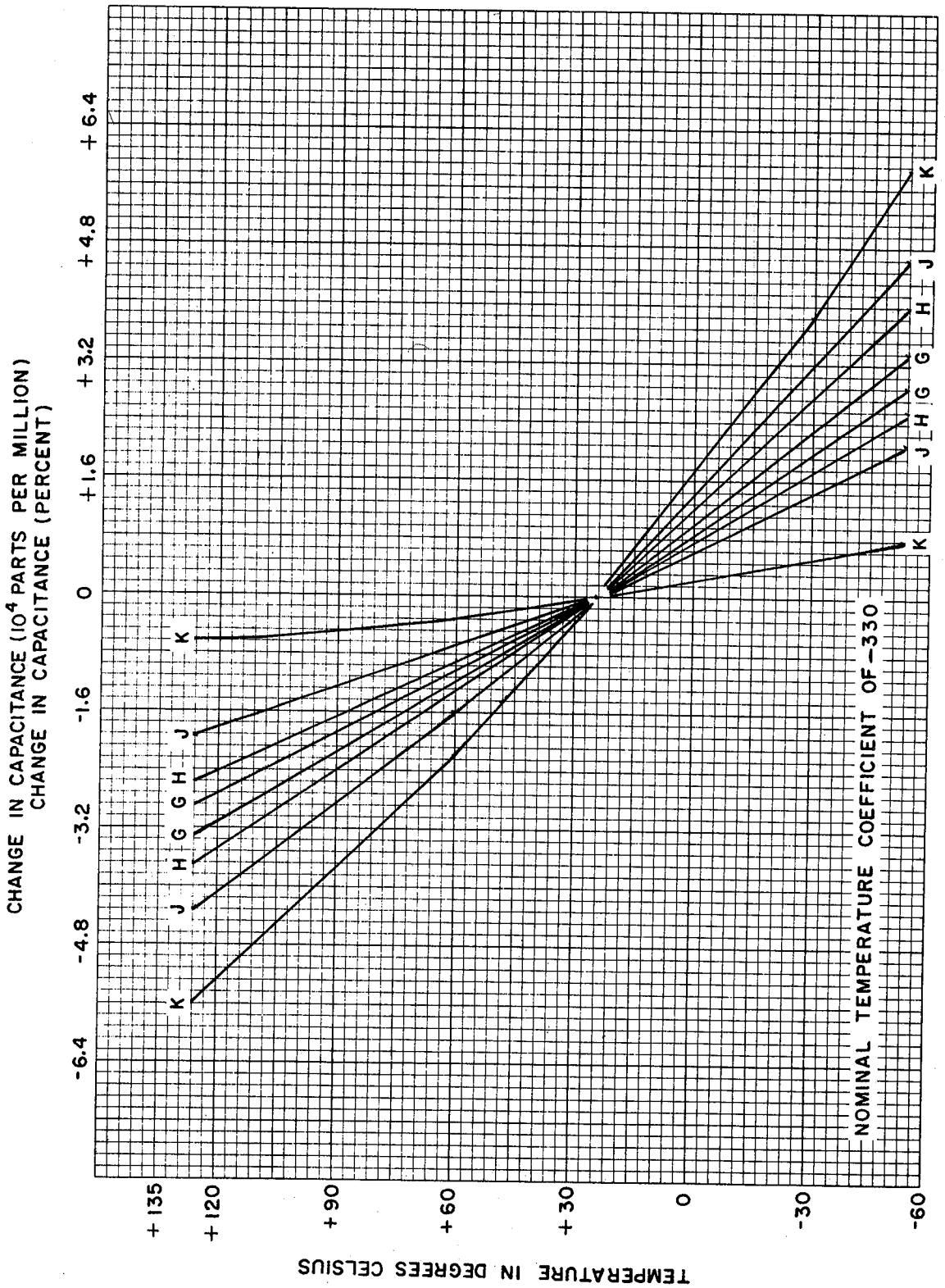


FIGURE 9. Tolerance envelopes for temperature coefficient S.

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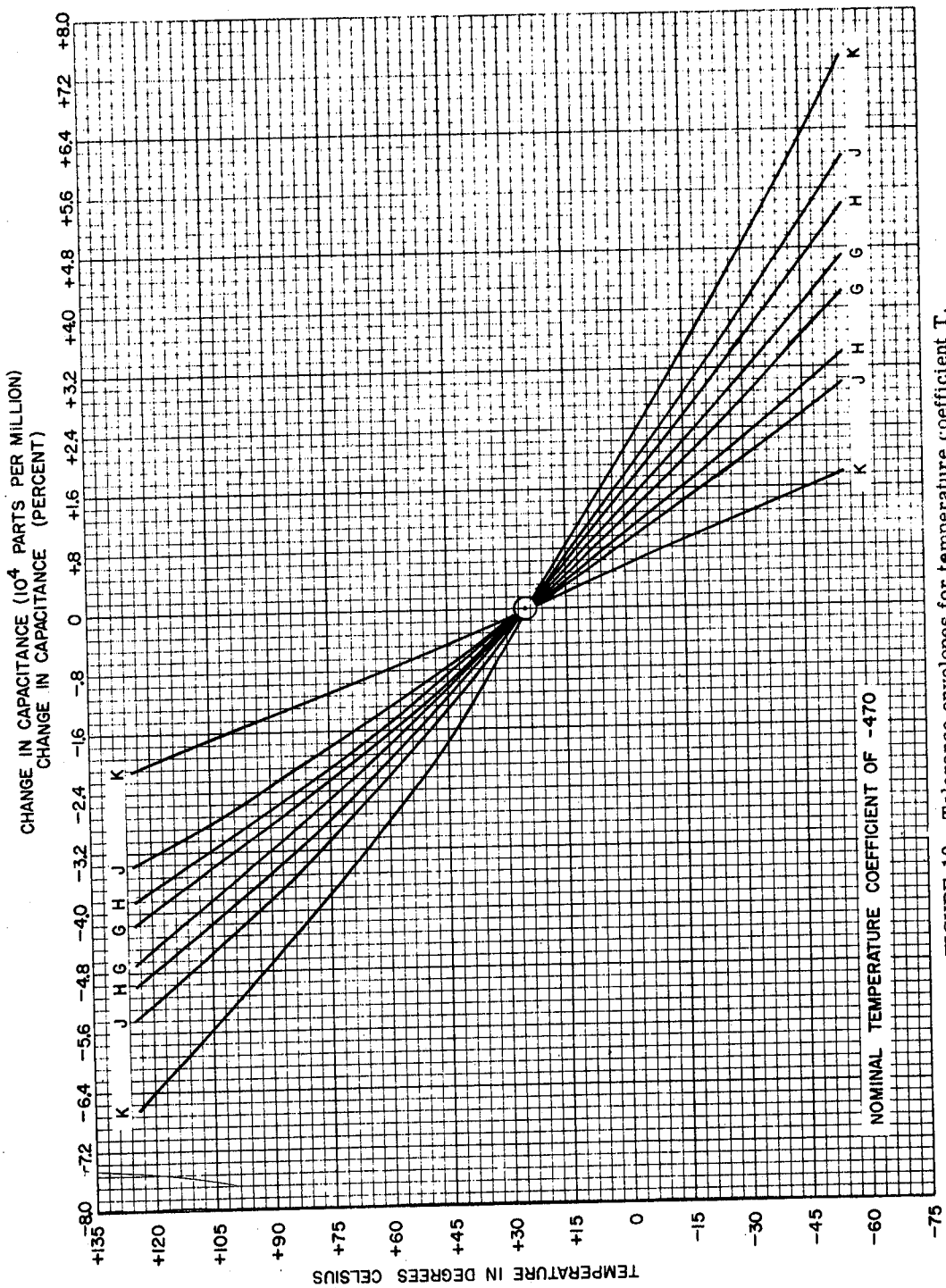


FIGURE 10. Tolerance envelopes for temperature coefficient T.

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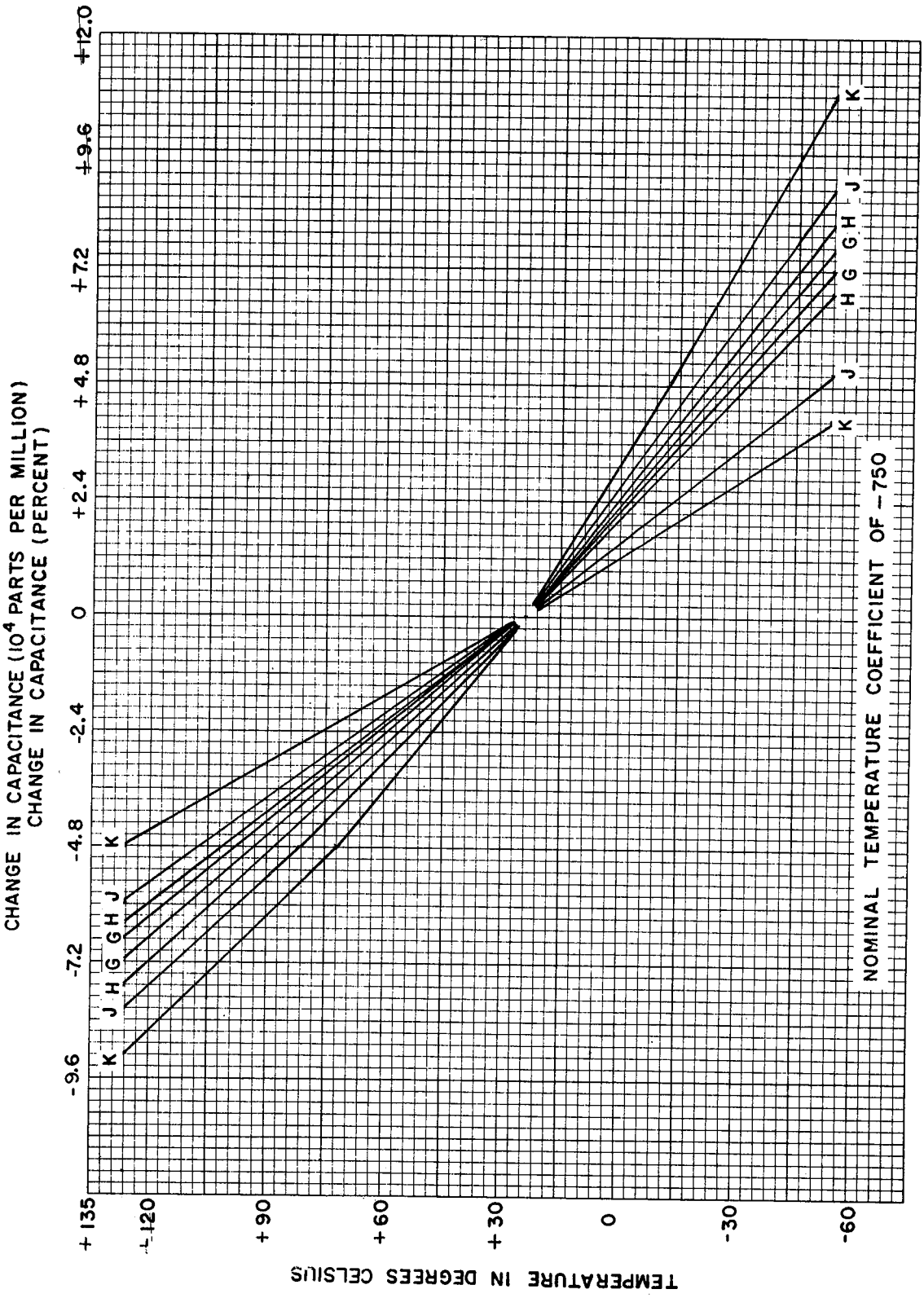


FIGURE 11. Tolerance envelope for temperature coefficient U.

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3.17 Thermal shock and immersion cycling. When tested as specified in 4.7,13, capacitors shall meet the following requirements:

Visual examination - - - - - -	There shall be no evidence of corrosion or mechanical damage.
Capacitance - --- - - - - - -	Shall change not more than ± (5 percent or 0.5 pF) (0.25 pF for values less than 10 pF), whichever is greater, from the initial value obtained when measured as specified in 4.7.4.
Dissipation factor or Q (as applicable, see 3.1) - -	Dissipation factor shall not exceed the initial requirement; Q shall be not less than the value shown on figure 1.
Dielectric withstanding voltage	Shall be as specified in 3.11.
Insulation resistance - - - -	Shall be not less than 50 percent of the initial requirement.

3.18 Terminal strength (direct load). When capacitors are tested as specified in 4.7.14, the terminals shall not loosen or rupture, and there shall be no other damage to the terminals or capacitor body.

3.19 Moisture resistance. When tested as specified in 4.7.15, capacitors shall meet the following requirements:

Visual examination - - - - - -	There shall be no evidence of corrosion or mechanical damage.
Capacitance - - - - - - - - -	Shall change not more than ± (5 percent or 0.5 pF) (0.25 pF for values less than 10 pF), whichever is greater, from the initial value obtained when measured as specified in 4.7.4.
Dielectric withstanding voltage	Shall be as specified in 3.11.
Insulation resistance - - - -	Shall be not less than 30 percent of the initial requirement.

3.20 Solderability (when specified, see 3.1). When capacitors are tested as specified in 4.7.16, the dipped surface of the leads shall be at least 95 percent covered with a new, smooth, solder coating. The remaining 5 percent may contain only small pinholes or rough spots; these shall not be concentrated in one area. Bare base metal where the solder dip failed to cover the original coating is an indication of poor solderability, and shall be cause for failure. In case of dispute, the percent of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area

3.21 Resistance to soldering heat (when specified, see 3.1). When tested as specified in 4.7.17, capacitors shall meet the following requirements:

Insulation resistance at 25°C - - - - -	Unless otherwise specified (see 3.1), not less than the initial 25°C requirement.
Capacitance - - - - - - - - - - - - -	Shall change not more than ±5 percent from the initial measured value.
Dissipation factor - - - - - - - - - -	Shall not exceed the initial requirement.

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3.22 Life (at elevated ambient temperature). When tested as specified in 4.7.18, capacitors shall meet the following requirements:

Visual examination	There shall be no evidence of corrosion or mechanical damage.
Capacitance	Shall change not more than ± (3 percent or 0.5 pF) (0.25 pF for values less than 10 pF), whichever is greater, from the initial value obtained when measured as specified in 4.7.4.
Dissipation factor or Q (as applicable, see 3.1)	Dissipation factor shall not exceed the initial requirement; Q shall be not less than the value shown on figure 1.
Insulation resistance	Shall be not less than 50 percent of the initial requirement.

3.23 Marking. Capacitors shall be marked as specified in 3.23.1 or 3.23.2, as applicable (see 3.1).

3.23.1 Method I. Marking shall be in accordance with method I of MIL-STD-1285. Unless otherwise specified (see 3.1), capacitors shall be marked with the "JAN" brand; type designation; date code and lot symbol; manufacturer's name or trademark; dc rated voltage; capacitance; and capacitance tolerance.

3.23.1.1 Type designation marking. There shall be no space between the symbols comprising the type designation. When specified (see 3.1), the type designation shall be divided between the characteristic and capacitance symbols. An ER part may be marked and furnished as a non-ER part, if produced on the same assembly line and provided it is subjected to and meets all the inspection requirements of the ER part.

3.23.1.2 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over, the certification marks "JAN" and "J", respectively, to indicate electrical equipment, namely, resistors, capacitors, electron tubes and the like, procured by, or manufactured for use by, or for the Government in accordance with standard Government specifications. Accordingly, capacitors procured to, and meeting all of, the criteria specified herein and in applicable specification sheets shall bear the certification mark "JAN", except that capacitors too small to bear the certification mark "JAN" shall bear the letter "J". Capacitors furnished under contracts which either permit or require deviation from the conditions or requirements specified herein and in applicable specification sheets shall not bear "JAN" or "J". In the event a capacitor sample fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all capacitors represented by the sample. The United States Government has obtained Certificate of Registration No. 504, 860 for the certification mark "JAN".

3.23.2 Method II. Marking shall be in accordance with method II of MIL-STD-1285; details and exceptions shall be as follows, and as specified (see 3.1):

a. The characteristic, capacitance, and capacitance tolerance shall be indicated on the capacitor by bands of color or spots of color, as applicable (see 3.1). The characteristic band (or spot) shall be wider (or larger) than any other band or spot.
b. The colors used shall be as specified in table IV. When the body color is the same as any of the band or spot colors, then either the body color, or the band or spot color, shall be differentiated by shade or gloss, or by some other means.

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TABLE IV. Color code.

Color	Characteristic 1/	Nominal capacitance (pF)		Capacitance tolerance	
		First and second significant figures	Multiplier 2/	For nominal capacitances greater than 10 pF	For nominal capacitances of 10 pF or smaller
				(±)	(±)
Black	C-	0	1	---	2.0% (G)
Brown	H-	1	10	1% (F)	---
Red	L-	2	100	2% (G)	0.25 pF (C)
Orange	P-	3	1,000	---	---
Yellow	R-	4	---	---	---
Green	S-	5	---	5% (J)	0.5 pF (D)
Blue	T-	6	---	---	---
Purple (violet)	U-	7	---	---	---
Gray	---	8	0.01	---	---
White	---	9	0.1	10% (K)	1.0% (F)
Gold	A-	---	---	---	---

1/ The characteristic is a two-letter symbol identifying the nominal temperature coefficient and the tolerance envelope for the temperature coefficient, respectively. However, the characteristic band or spot identifies only the nominal temperature coefficient.

2/ The multiplier is the factor by which the two significant figures are multiplied to yield the nominal capacitance (see 1.2.1.3). The lowest possible numerical multiplier shall be used to avoid alternate coding; for example, 0.5 pF should be green, black, gray - NOT black, green, white.

3.24 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, chips, and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Reliability assurance program (applicable to ER parts only). A reliability assurance program shall be established and maintained in accordance with MIL-STD-790, with the following exceptions:

a. The following shall apply in lieu of paragraph 5.2.7(j): The manufacturer shall as a minimum be able to identify the time period during which the final production operation was performed on each item of product prior to final test. The date or lot code marked on each part shall be identified to an inspection lot.
b. Paragraph 5.2.11.3 shall not apply.

Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Verification of qualification (see 4.5).
c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions and methods.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be made in accordance with the "GENERAL REQUIREMENTS" of MIL-STD-202 except relative humidity shall not exceed 75 percent. Accuracy of all test voltage measurements shall be within ±2.0 percent of the specified voltage.

4.3.2 Methods.

4.3.2.1 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25° ±3°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the specified test voltage.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3), on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number of capacitors to be specified to qualification inspection shall be as specified in table V and in the appendix of this specification. Each capacitor style shall be qualified separately.

4.4.2 Inspection routine. The sample shall be subjected to the qualification inspection specified in table V, in the order shown. ER styles shall be subjected to groups I through VI; non-ER styles, active for new design, shall be subjected to groups II through VI; and non-ER styles, inactive for new design, shall be subjected to groups II, III, IV, and VI. For ER styles, all samples shall be subjected to group I, no failures permitted; group II, 1 failure permitted; remaining samples shall be divided as shown in table V for groups III through VI, respectively. For non-ER styles, active for new design, samples shall be subjected to group II, and then divided as shown in table V for groups III through VI, respectively. For non-ER styles, inactive for new design, samples shall be subjected to group II, and then divided as shown in table V for groups III, IV, and VI, respectively.

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4.4.3 Failures. Failures in excess of those allowed in table V shall be cause for refusal to grant qualification approval.

4.4.4 Failure rate (FR) qualification (ER styles only). Failure rate qualification shall be in accordance with the general and detailed requirements of MIL-STD-690 with the following details:

a. Procedure I - Qualification at the initial failure rate level. Level 1.0 percent/1,000 hours "M" of FRSP-90 shall apply. Sample units which have been subjected to the qualification inspection specified in group VI, table V shall be continued on test as specified in 4.7.18.2.2.
b. Procedure II - Extension of qualification to lower failure rate levels. To extend qualification to the 0.01 percent/1,000 hours "R" and 0.001 percent/1,000 hours "S" failure rate levels, data from two or more styles of similar construction may be combined.
c. Procedure III - Maintenance of failure rate level qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification.

4.4.5 Retention of qualification (non-ER styles only). To retain qualification, the contractor shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
b. The results of tests performed for periodic inspection (group B), including the number and mode of failures. The test report shall include results of all periodic inspection tests performed and completed during the 6-month period. If the test results indicate nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 6-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during three consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each style to testing in accordance with the qualification inspection requirements.

Retention of qualification of ER style will also retain qualification of similar non-ER style; e.g., CCR75 and CC75.

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TABLE V. Qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of failures permitted 1/
2/ Group I				
Thermal shock and voltage conditioning (ER parts only)	3.6	4.7.2	55	0
2/ Group II				
Visual and mechanical inspection: Material, design and construction, marking 3/, and workmanship	3.1, 3.4, 3.5, 3.23, and 3.24	4.7.1	4/ 43 (Non-ER) or 55 (ER)	1
RF current (when specified, see 3.1)	3.7	4.7.3		
Capacitance	3.8	4.7.4		
Dissipation factor (when specified, see 3.1)	3.9	4.7.5		
Quality factor (Q) (when specified, see 3.1)	3.10	4.7.6		
Dielectric withstanding voltage	3.11	4.7.7		
Barometric pressure	3.12	4.7.8		
Insulation resistance	3.13	4.7.9		
Group III				
Temperature coefficient and capacitance drift	3.14	4.7.10.1	12	1
Shock (specified pulse)	3.15	4.7.11		
Vibration, high frequency	3.16	4.7.12		
Thermal shock and immersion cycling	3.17	4.7.13		
Group IV				
Terminal strength (direct load)	3.18	4.7.14	6	1
Moisture resistance	3.19	4.7.15		
5/ Group V				
Solderability (when specified, see 3.1)	3.20	4.7.16	6	1
Resistance to soldering heat (when specified, see 3.1)	3.21	4.7.17	6	
Group VI				
Life (at elevated ambient temperature)	3.22	4.7.18.1	12 (Non-ER) or 24 (ER)	1

1/ A sample unit having one or more defects shall be considered as a single failure.
2/ All examinations and tests are nondestructive.
3/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect marking. Provisions to reject unmarked parts will also be made.
4/ One additional sample unit is included in each sample to permit substitution for the failure allowed in group II (43 sample units for inactive for new design styles; 55 sample units for active for new design styles).
5/ Applicable to active for new design styles only.

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4.5 Verification of qualification (ER styles only). Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspections and, where applicable, extended failure rate test data, in the form of a verification of qualification report, and forward it to the qualifying activity as the basis of continued qualification approval. In addition to the periodic submission of failure rate test data, the manufacturer shall immediately notify the qualifying activity whenever the failure rate data indicates that the manufacturer has failed to maintain his qualified failure rate level. Continuation shall be based on evidence that, over the 6-month period, the following has been met:

a. Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.
b. The manufacturer has not modified the design of the item.
c. The specification requirements for the item have not been amended so far as to affect the character of the item.
d. Lot rejection for group A inspection does not exceed 10 percent or one lot, whichever is greater.
e. The data from group A quality conformance life test (failure rate level 1.0 percent/1,000 hours only) substantiates that the failure rate level is being maintained.
f. The requirements for group B inspection are met.
g. The records of all failure rate tests combined substantiate that the 1.0 percent/1,000 hours, or 0.1 percent/1,000 hours "P" failure rate levels have been maintained or that the manufacturer continues to meet the 0.01 percent/1,000 hours, or 0.001 percent/1,000 hours failure rate level for which qualified although the total component hours of testing does not, as yet, meet the requirements of 4.4.4c.

If group B test requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the forwarding of the verification of qualification report may be delayed until within 30 days after completion of retesting of the periodic quality conformance tests. In this case, the qualifying activity shall be notified of this condition within the time the original verification of qualification report was due. All reports shall be certified by the responsible company official. The qualifying activity shall be contacted for the report format.

4.5.1 Records. Test records shall be in accordance with the format in MIL-STD-690.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection (for FR levels 1.0, 0.1, 0.01 and 0.001 percent/1,000 hours).

4.6.1.1 Inspection lot. An inspection lot shall consist of all capacitors in one or more styles, produced under essentially the same conditions, and offered for inspection at the same time. ER parts shall be kept separate from non-ER parts. The sample selected from the lot shall be representative of the styles in the lot. The styles may be grouped as follows:

Group	Styles
1	CCR05, CCR06, CCR07, and CCR08.
2	CCR13, CCR14, CCR15, CCR16, CCR17, and CCR18.
3	CCR75, CCR76, CCR77, CCR78, and CCR79.
4	All styles "inactive for new design" may be combined.

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TABLE VI. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Non-ER parts AQL (percent defective)		ER parts LQ (limiting quality) And AQL	
Subgroup 1						
Thermal shock and voltage conditioning (ER parts only) - - -	3.6	4.7.2	Not applicable		100% inspection	
Subgroup 2			Major	Minor	Major	Minor
Visual and mechanical inspection: Material, design and construction, marking 1/, and workmanship - - -	3.1, 3.4, 3.5, 3.23, and 3.24	4.7.1	0.65	2.5	0.65 (AQL) 4.8 (LQ)	2.5 (AQL) 13 (LQ)
RF current (when specified, see 3.1) - - - - - - - - - - - - -	3.7	4.7.3	0.65	---	Not applicable	
Capacitance - - - - - - - - - - - - -	3.8	4.7.4				
Dissipation factor (when specified, see 3.1) - - - -	3.9	4.7.5				
Quality factor (Q) (when specified, see 3.1) - - - -	3.10	4.7.6				
Dielectric withstanding voltage - -	3.11	4.7.7				
Insulation resistance - - - - - - -	3.13	4.7.9				
Life (at elevated ambient temperature) (performance check) - - - -	3.22	4.7.18.2.1	2.5	---	2.5 (AQL)	---

1/ Marking defects are based on visual examination only.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table VI, in the order shown.

4.6.1.2.1 Sampling plan. Subgroup 1 tests shall be performed on each ER capacitor offered for inspection. Statistical sampling and inspection for subgroup 2 shall be in accordance with MIL-STD-105. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.2.2 Manufacturer's production inspection (ER parts only). If the manufacturer performs tests similar to those specified in subgroup 1, table VI, as the final step of his production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria shall be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 1.
b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.
c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.
d. The lot rejection criteria is the same or more stringent than that specified herein.
e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. This data shall be provided as part of the evaluation required for MIL-STD-790. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.
f. Once approved, the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.6.1.2.3 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. If during the 100-percent inspection of subgroup 1 (ER parts only), screening requires that over 5 percent of the capacitors be discarded, the lot shall be rejected and shall not be resubmitted.

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4.6.2 Periodic (group B) inspection. Periodic inspection shall consist of group B. Except where the results of these inspections show noncompliance with the applicable requirements (see 6.7), delivery of products which have passed group A shall not be delayed pending the results of these periodic inspections.

4.6.2.1 Group B inspection. Group B inspection shall consist of the inspections specified in table VII, in the order shown, and shall be performed on sample units selected from lots that have passed group A inspection. Test data obtained shall be reviewed as part of the complete retention of qualification.

TABLE VII. Group B inspection.

Test	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of defectives permitted 1/
Every month				
Barometric pressure	3.12	4.7.8	12	0
Temperature coefficient and capacitance drift	3.14	4.7.10.2		
2/ Every 2 months				
3/ Subgroup 1				
Shock (specified pulse	3.15	4.7.11	12	1
Vibration, high frequency	3.16	4.7.12		
Thermal shock and immersion cycling	3.17	4.7.13		1
Subgroup 2				
Terminal strength (direct load)	3.18	4.7.14	12	1
Moisture resistance	3.19	4.7.15		
4/ Subgroup 3				
Solderability (when specified, see 3.1)	3.20	4.7.16	6	1
Resistance to soldering heat (when specified, see 3.1)	3.21	4.7.17	6	1
Every 4 months				
Life (at elevated ambient temperature) (continuation) 5/	3.22	4.7.18.2.2	12	1

1/ A sample unit having one or more defects shall be considered as a single failure.
2/ Subgroups 1 and 2 shall be checked during alternate bimonthly periods.
3/ Subgroup 1 tests may be performed on sample units that have been subjected to and have passed the monthly inspection, when these sampling periods coincide.
4/ Applicable to new design styles (ER and non-ER) only.
5/ Sample units to be inspected shall be selected from sample units which have been subjected to the performance-check test (see table VI) and shall then be tested for an additional period of $750\ ^{+48}_{-0}$ (non-ER) or $1{,}750\ ^{+96}_{-0}$ (ER) hours; the continuation test shall be started no later than 96 hours after completion of the performance-check test.

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4.6.2.1.1 Sampling plan.

4.6.2.1.1.1 Every month. Every month, sample units of each temperature coefficient and the tightest temperature-coefficient tolerance in each temperature coefficient shall be selected and subjected to the applicable tests specified in table VII.

4.6.2.1.1.2 Every 2 months. Once each 2 calendar months, sample units of the same type designation shall be selected and subjected to the tests of subgroups 1, 2, and 3 of table VII. When each sample is selected, the Government shall review all selections made within the preceding 2 years in order to assure that all styles have been drawn into the testing program.

4.6.2.1.1.3 Every 4 months. Every 4 months, sample units of the highest capacitance value of each style produced shall be selected and subjected to the test specified in table VII. These capacitors may be in any capacitance tolerance.

4.6.2.1.2 Defectives. If the number of defectives exceeds the number allowed in table VII, the sample shall be considered to have failed.

4.6.2.1.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract.

4.6.3 Inspection of packaging. The sampling and inspection of the preservation-packaging, packing, and container marking for shipment and storage shall be in accordance with the requirements of MIL-C-39028.

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. Capacitors shall be examined to verify that the material, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.23, and 3.24).

4.7.2 Thermal shock and voltage conditioning (ER parts only) (see 3.6). Capacitors shall be subjected to the tests of 4.7.2.1 and 4.7.2.2.

4.7.2.1 Thermal shock. Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

a. Test condition - B.
b. Measurements before and after cycling - Not applicable.

4.7.2.2 Voltage conditioning. Voltage conditioning shall be started immediately after completion of the temperature cycling test. The voltage conditioning shall consist of applying twice the rated voltage to the unit at 125° $^{+4}_{-0}$°C for 100 ±4 hours. After completion of the exposure period and while at high temperature, the insulation resistance shall be measured. (This measurement will be considered the initial measurement of insulation resistance at high test temperature.) At the option of the manufacturer, the units may be immediately transferred (period of transfer not to exceed 15 minutes) to another chamber maintained at the same temperature for the purpose of measuring insulation resistance. After stabilization at room temperature, the capacitance, dissipation factor, and insulation resistance shall then be measured as specified in 4.7.4, 4.7.5, and 4.7.9.1, respectively.

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4.7.3 RF current (when specified, see 3.1) (see 3.7). Capacitors shall have an RF voltage applied at the test frequency specified, (see 3.1). The RF current and the temperature of the capacitor shall be monitored. The temperature of the capacitor shall not rise more than 30°C above a 25°C ambient, in still air, with the rated RF current applied (see 3.1). The capacitors shall be mounted in such a way that the substrate shall not act as a heat sink.

4.7.4 Capacitance (see 3.8). Capacitance shall be measured in accordance with method 305 of MIL-STD-202. The following details and exception shall apply:

a. Test frequency - 1 megahertz (MHz) ±100 kilohertz (kHz), when the capacitance is 1,000 pF and smaller; and 1 kHz ± 100 Hz, when the capacitance is greater than 1,000 pF. At the option of the manufacturer, capacitance measurements may be made at any frequency from 1 kHz to 1 MHz and referred to measurements at 1 MHz and 1 kHz, as applicable.
b. Limit of accuracy - The accuracy of measurement shall be 1/3 of the nominal capacitance tolerance or 2 percent, whichever is smaller; however, the accuracy need not be better than 0.1 pF.
c. Test jig for measuring capacitance values less than 10 pF - Shall be a guarded three terminal type, or equal.

4.7.5 Dissipation factor (when specified, see 3.1) (see 3.9). The dissipation factor shall be measured with a capacitance bridge or other suitable method at the frequency specified in 4.7.4a. The voltage shall be 1.0 ±0.2 volts rms. The inherent accuracy of the measurement shall be ±2 percent of the reading plus 0.1 percent dissipation factor (absolute). Suitable measurement techniques shall be used to minimize errors due to the connections between the measuring apparatus and the capacitor.

4.7.6 Quality factor (Q) (when specified, see 3.1) (see 3.10). Capacitors shall be tested in accordance with method 306 of MIL-STD-202. The following detail shall apply:

a. Test frequency - As specified in 4.7.4a.

4.7.7 Dielectric withstanding voltage (see 3.11). Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Magnitude and nature of test voltage - 300 percent of dc rated voltage specified (see 3.1).
b. Duration of application of test voltage - 5 ±1 seconds.
c. Points of application of test voltage - Between capacitor-element terminals.
d. Limiting value of surge current - Shall not exceed 50 milliamperes (mA).
e. Examinations after test - Capacitors shall be examined for evidence of damage and breakdown.

4.7.7.1 Body insulation (when specified, see 3.1). Capacitors shall be tested in accordance with one of the following, as specified (see 3.1).

a. Test I - Capacitors shall be placed in the trough of a V-block which shall extend beyond the ends of the body of the capacitor. A dc potential of 1,300 volts shall then be applied between the two leads connected together and the V-block for a period of 5 ±1 seconds. The surge current shall not exceed 50 mA.

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b. Test II - Capacitors shall be wrapped with a conductive tape or foil so that the conductive tape or foil shall be not less than 0.0625 inch and not more than 0.125 inch away from the lead wires. A dc potential of 1,300 volts shall be applied between the two leads connected together and the tape or foil for a period of 5 +1 seconds. The test circuit shall be so arranged that the surge current does not exceed 50 mA.

After the test, capacitors shall be examined for evidence of damage and breakdown.

4.7.8 Barometric pressure (see 3.12). Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting - Capacitors shall be mounted by suitably clamping their leads.
b. Test-condition - B.
c. Tests during subjection to reduced pressure - 100 percent of the dc rated voltage (see 3.1) shall be applied between capacitor-element terminals for a period of 5 +1 seconds. The surge current shall not exceed 50 mA.

4.7.9 Insulation resistance (see 3.13).

4.7.9.1 At +25°C. Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test potential - Rated voltage (see 3.1).
b. Points of measurement - Between capacitor-element terminals.
c. Electrification time - 2 minutes +5 seconds, except during quality conformance inspection when the electrification time may be the time required by the capacitor to reach full charge, provided it does not exceed 2 minutes. Voltage shall be applied through a resistor which will limit the charging current from 30 to 50 mA.

4.7.9.2 At high ambient temperature. Capacitors shall be subjected to the appli- high ambient temperature (see 3.1) for a period of time sufficient to reach thermal stability, and the insulation resistance shall then be measured as specified in 4.7.9.1

4.7.10 Temperature coefficient and capacitance drift (see 3.14).

4.7.10.1 For qualification inspection.

4.7.10.1.1 Temperature coefficient. Capacitance measurements shall be made as specified in 4.7.4 and at the temperatures specified in table VIII.

4.7.10.1.2 Capacitance drift. Capacitance drift shall be computed by dividing the greatest single difference between any two of the three capacitance values recorded at 25°C by the value determined at the reference temperature. (See table VIII.)

4.7.10.2 For quality conformance inspection.

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TABLE VIII. Temperature-coefficient-and-capacitance-drift cycle.

Sequence	Temperature
	°C
(a)	$+25^{+0}_{-2}$
(b)	-55^{+0}_{-2}
(c)	-40^{+0}_{-2}
(d)	-10^{+0}_{-2}
(e)	1/ $+25^{+0}_{-2}$
(f)	$+45^{+2}_{-0}$
(g)	$+65^{+2}_{-0}$
(h)	$+85^{+2}_{-0}$
(i)	2/ $+105^{+2}_{-0}$
(j)	2/ $+125^{+2}_{-0}$
(k)	$+25^{+0}_{-2}$

1/ Reference temperature.
2/ Applicable only to 125°C capacitors.

The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals at that temperature indicate no change in capacitance. The accuracy of measurement of capacitance increments shall be within ±0.005 pF. For capacitors with values of 20 pF or less, the temperature-coefficient tolerance shall be as specified (see 3.1).

4.7.10.2.1 Temperature coefficient. Capacitance measurements shall be made as specified in 4.7.10.1.1, except that measurements shall be made only at $+25^{\circ}{}^{+0^{\circ}}_{-2^{\circ}}$C, $-55^{\circ}{}^{+0^{\circ}}_{-2^{\circ}}$C, $+25^{\circ}{}^{+0^{\circ}}_{-2^{\circ}}$C (reference temperature), $+85^{\circ}{}^{+2^{\circ}}_{-0^{\circ}}$C, $+125^{\circ}{}^{+2^{\circ}}_{-0^{\circ}}$C (if applicable, see 3.1), and $+25^{\circ}{}^{+0^{\circ}}_{-2^{\circ}}$C, respectively, and the values so determined shall be checked against the known shape of the curve for the dielectric material used, as determined in more complete tests.

4.7.10.2.2 Capacitance drift. Capacitance drift shall be computed as specified in 4.7.10.1.2.

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4.7.11 Shock (specified pulse) (see 3.15). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details and exception shall apply:

a. Special mounting means - Capacitors shall be rigidly mounted on a mounting fixture by the body; uninsulated styles shall have provision for insulating the body. Leads shall be secured to rigidly-supported terminals, so spaced that the length of each lead from the capacitor is approximately 3/8 inch when measured from the edge of the supporting terminal. Leads shall be within 30 degrees of being parallel; each lead of disk types shall be in the plane of the flat surface from which it extends. When securing leads, care shall be taken to avoid pinching the leads.
b. Number and direction of applied shocks - Thirty impact shocks, 10 blows in each of three mutually perpendicular planes (five blows in each of two directions). The test planes shall be parallel to the major axes of the capacitor body (see figure 12).
c. Test-condition - I.
d. Electrical-load conditions - During the test, a potential of 125 percent of the dc rated voltage (see 3.1) shall be applied between the terminals of the capacitor.
e. Measurements during shock - During the test, observations shall be made for momentary or intermittent arcing. A cathode-ray oscilloscope, or other comparable means, shall be used as an indicating device in determining any electrical failures.
f. Examinations after test - Capacitors shall be visually examined for evidence of breakdown, arcing, and mechanical damage.

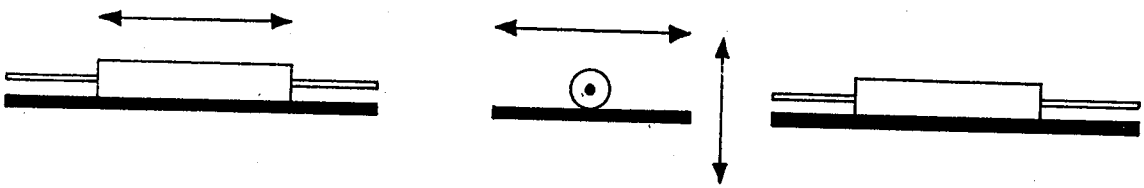


FIGURE 12. Planar diagram for shock test.

4.7.12 Vibration, high frequency (see 3.16). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exception shall apply:

a. Mounting of specimens - Capacitors shall be mounted as specified in 4.7.11a. The mounting fixture shall be so constructed as to preclude any resonances within the test range. An examination of the mounting fixture shall be made on a vibrator. If any resonant frequencies are observed, adequate steps must be taken to damp the structure.
b. Electrical-load conditions - As specified in 4.7.11d.
c. Test-condition -B.
d. Duration and direction of motion - 4 hours in each of two mutually perpendicular planes (total of 8 hours), one parallel and the other perpendicular to the major axis.
e. Measurements during vibration - As specified in 4.7.11e.
f. Examination and measurements after vibration - After the test, capacitors shall be visually examined for evidence of mechanical damage; the capacitance, dissipation factor or Q (as specified, see 3.1), dielectric withstanding voltage, and insulation resistance shall then be measured as specified in 4.7.4, 4.7.5, or 4.7.6, 4.7.7 (and 4.7.7.1, if applicable), and 4.7.9.1, respectively.

4.7.13 Thermal shock and immersion cycling (see 3.17).

4.7.13.1 Thermal shock. Capacitors shall be tested in accordance with method 107, test condition A, of MIL-STD-202, except that the maximum temperature shall be $85°^{+3°}_{-0°}$ or $125°^{+3°}_{-0°}$, as applicable (see 3.1), and no measurements shall be made before and after cycling.

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4.7.13.2 Immersion cycling. Following thermal shock, capacitors shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

a. Test-condition - B.
b. Examinations and measurements after final cycle - Capacitors shall be visually examined for evidence of corrosion and mechanical damage; the capacitance, dissipation factor or Q (as specified, see 3.1), dielectric withstanding voltage, and insulation resistance shall then be measured as specified in 4.7.4, 4.7.5 or 4.7.6, 4.7.7 (and 4.7.7.1, if applicable), and 4.7.9.1, respectively.

4.7.14 Terminal strength (direct load) (see 3.18). Capacitors shall be held by one terminal, and the load shall be gradually applied in the direction of the leads until the applied load reaches 5 pounds. The 5-pound pull shall be applied for at least 5 seconds. The capacitors shall then be examined for evidence of loosening or rupture of the terminals, and any other damage to the terminals and capacitor body.

4.7.15 Moisture resistance (see 3.19). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following detail and exceptions shall apply:

a. Initial measurements - Not applicable.
b. Polarization (unless otherwise specified, see 3.1) - During the first 10 cycles only, a dc potential of 100 volts or rated voltage, whichever is less, shall be applied across the capacitor terminals. Once each day a check shall be performed to determine whether a capacitor has shorted.
c. Examinations and final measurement - Upon completion of step 6 of the final cycle, capacitors shall be visually examined for evidence of mechanical damage and obliteration of marking; capacitance, dielectric withstanding voltage, and insulation resistance shall then be measured as specified in 4.7.4, 4.7.7 (and 4.7.7.1, if applicable), and 4.7.9.1, respectively.

4.7.16 Solderability (when specified, see 3.1) (see 3.20). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following detail shall apply:

a. Number of terminations to be tested - Two.

4.7.17 Resistance to soldering heat (when specified, see 3.1) (see 3.21). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Depth of immersion in the molten solder - To a minimum of 0.050 inch from the capacitor body (the example shown on figure 13 is applicable to all terminal types).
b. Test condition - D, except that the immersion duration shall be 20 ±1 seconds.
c. Cooling time prior to measurement after test - 10 ±1 minutes, unless otherwise specified (see 3.1).
d. Measurements after test - Capacitance, dissipation factor, and insulation resistance shall be measured as specified in 4.7.4, 4.7.5, and 4.7.9.1, respectively.

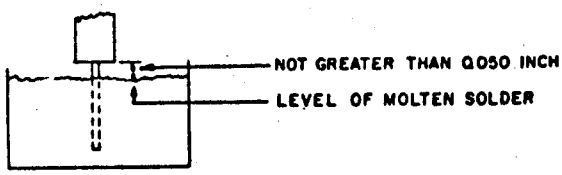


FIGURE 13. Example of axial wire-lead depth of immersion in molten solder.

4.7.18 Life (at elevated ambient temperature) (see 3.22).

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4.7.18.1 For qualification inspection. Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Distance of temperature measurements from specimens, in inches - Not applicable.
b. Test temperature and tolerance - At the applicable elevated temperature, ±3°C (see 3.1).
c. Operating conditions - One of the following, as specified (see 3.1).

Condition 1: Capacitors shall be subjected to a dc potential equal to 150 percent of the dc rated voltage on which an alternating-current rms potential (60 ±2 Hz) equal to 50 percent of dc rated voltage is superimposed.

Condition 2: Capacitors shall be subjected to a dc potential equal to 200 percent of the dc rated voltage. Current shall be limited from 30 to 50 mA.

d. Test-condition - D, ±12 hours (non-ER) or F, ±24 hours (ER).
e. Examination and measurements after exposure - Capacitors shall be returned to the inspection conditions specified in 4.3, and shall be visually examined for evidence of corrosion and mechanical damage; the capacitance, dissipation factor or Q (as specified, see 3.1), and insulation resistance shall then be measured as specified in 4.7.4, 4.7.5, or 4.7.6, and 4.7.9.1, respectively.

4.7.18.2 For quality conformance inspection.

4.7.18.2.1 Performance check. Capacitors shall be tested as specified in 4.7.18.1, except that test-condition B, ±4 hours shall be used.

4.7.18.2.2 Continuation. Capacitors which have been subjected to the performance-check test specified in 4.7.18.2.1 shall be tested for an additional period of 750^{+48}_{-0} (non-ER parts) or 1,750 $^{+96}_{-0}$ (ER parts) hours in accordance with 4.7.18.1, with voltage applied with the same polarity as when subjected to the performance-check test. (See table VII.)

5. PACKAGING.

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-39028.

6. NOTES

6.1 Intended use. These capacitors are designed for use primarily where compensation is needed to counteract reactive changes, caused by temperature variations, in other circuit components. However, they can be used in any precision-type circuit where their characteristics are suitable. Ceramic capacitors are substantially smaller than paper or mica units of the same capacitance and voltage rating. They can be used where mica or paper capacitors have too wide a capacitance tolerance. The lead placement makes ceramic capacitors suitable for printed-circuit use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet, and the complete type designation (see 1.2.1 and 3.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the qualified products list is Aeronautical Systems Division, ASD/ENYES, Wright-Patterson AFB, Ohio 45433; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444.

6.4 Quality factor (Q). Q is defined as the value of the tangent of the angle between the voltage and current vectors for the capacitor at the specified frequency, and is expressed as follows:

$$Q = \frac{1}{R\omega C}$$

Where:

R = equivalent series resistance (in ohms).

ω = 2π times the frequency (in Hertz).

C = capacitance (in farads).

6.5 Characteristic. To facilitate use of the curves on figures 3 to 11 inclusive, maximum and minimum readings at -55°, -40°, -10°, +45°, +65°, +85°, and +125°C, for each characteristic are given in table IX.

6.6 Selection and use information. Equipment designers should refer to MIL-STD-198, "Capacitors, Selection and Use Of", for standard capacitor types and selected values chosen from this specification. MIL-STD-198 provides a selection of standard capacitors for new equipment design.

6.7 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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TABLE IX. Points read from tolerance envelopes on figures 3 to 11 inclusive.

Characteristic	Permissible capacitance change from capacitance at 25°C (parts per million)													
	At -55°C		At -40°C		At -10°C		At +45°C		At +65°C		At +85°C		At +125°C	
	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
AH	−2,100	−12,500	−1,800	−10,300	−1,000	−5,600	3,000	700	6,300	1,500	9,600	2,400	16,200	4,100
AJ	3,300	−17,300	2,600	−14,000	1,300	−7,700	4,200	−500	8,800	−900	13,200	−1,200	21,400	−1,500
AK	16,300	−27,500	13,000	−22,400	6,700	−12,200	6,500	−3,300	13,800	−6,300	21,000	−9,000	34,100	−13,500
CF	2,800	−900	2,000	−800	900	−500	300	−300	600	−600	900	−900	1,500	−1,500
CG	4,300	−2,200	3,400	−1,900	1,700	−1,100	600	−700	1,100	−1,400	1,700	−1,800	3,000	−3,000
CH	7,300	−4,400	5,800	−3,700	2,900	−2,000	1,300	−1,500	2,400	−2,500	3,600	−3,600	6,000	−6,000
CJ	13,300	−9,300	10,500	−7,600	5,400	−4,200	2,300	−2,700	4,700	−5,100	7,200	−7,200	12,000	−12,000
CK	26,100	−19,700	20,800	−16,100	10,700	−8,600	5,000	−5,500	10,000	−10,400	15,000	−15,000	25,000	−25,000
HF	5,800	1,800	4,600	1,400	2,300	700	−300	−1,100	−600	−2,000	−800	−2,800	−1,000	−4,100
HG	7,300	300	5,900	200	2,900	100	0	−1,400	0	−2,600	0	−3,600	0	−5,200
HH	10,300	−2,100	8,200	−1,800	4,200	−1,000	500	−2,100	1,200	−3,800	1,800	−5,400	3,200	−8,300
HJ	16,200	−6,900	13,000	−5,600	6,700	−3,000	1,700	−3,300	3,500	−6,300	5,300	−9,000	8,900	−13,300
HK	29,300	−17,500	23,400	−14,200	12,100	−7,600	4,400	−6,000	8,800	−11,800	13,200	−16,800	21,200	−25,500
LF	10,400	7,700	8,000	6,000	4,400	3,400	−1,500	−1,800	−3,100	−4,300	−4,400	−6,200	−6,700	−9,700
LG	12,200	5,300	9,800	4,200	5,000	2,200	−1,100	−2,500	−2,100	−4,700	−3,000	−6,600	−5,200	−11,200
LH	15,200	2,400	12,100	1,900	6,200	900	−500	−3,100	−900	−5,900	−1,200	−8,400	−2,200	−14,200
LJ	21,200	−2,900	16,900	−2,400	8,700	−1,300	700	−4,500	1,600	−8,400	2,400	−12,000	3,800	−20,200
LK	34,300	−13,300	27,300	−10,900	14,200	−5,900	3,300	−7,100	6,700	−13,800	−10,000	−19,800	16,800	−33,200
PF	16,000	13,600	12,800	10,800	6,700	5,600	−2,800	−3,400	−5,500	−6,700	−8,100	−9,900	−12,600	−15,700
PG	19,300	12,400	15,500	10,200	8,000	5,100	−2,400	−3,800	−4,800	−7,400	−7,200	−10,900	−11,000	−17,600
PH	23,300	9,500	18,000	7,900	9,300	3,900	−1,800	−4,600	−3,600	−8,600	−5,500	−12,600	−8,800	−20,200
PJ	28,000	3,400	22,500	2,700	11,500	1,300	−600	−5,700	−1,200	−11,200	−1,800	−16,300	−3,000	−25,600
PK	41,000	−7,500	33,000	−6,400	17,100	−3,500	1,800	−8,600	3,800	−16,500	6,000	−24,100	10,700	−39,300
RF	22,200	19,800	18,000	16,000	9,100	8,200	−4,800	−5,400	−9,000	−10,200	−13,000	−14,800	−20,200	−23,200
RG	24,000	18,700	19,100	14,900	9,800	7,700	−4,500	−5,700	−8,400	−10,800	−12,100	−15,700	−18,700	−24,500
RH	29,100	15,500	23,000	12,000	12,200	6,100	−3,100	−6,200	−6,200	−11,900	−9,500	−16,800	−15,700	−25,700
RJ	35,300	10,700	28,000	8,200	14,600	3,900	−1,900	−7,400	−3,900	−14,000	−6,000	−20,500	−10,500	−32,200
RK	48,000	−2,000	38,000	−1,700	19,700	−1,000	600	−10,000	1,300	−19,500	1,800	−28,400	2,700	−44,400
SG	34,400	29,600	28,000	24,100	14,700	12,600	−6,400	−7,600	−12,300	−14,700	−18,100	−21,700	−28,800	−34,600
SH	40,800	26,000	31,900	20,300	16,600	10,000	−5,300	−8,300	−11,300	−16,500	−16,000	−23,600	−25,600	−37,600
SJ	46,200	21,500	36,400	17,200	18,900	9,000	−4,500	−9,800	−9,000	−18,800	−12,600	−26,900	−19,600	−43,600
SK	59,100	8,300	47,600	6,800	24,200	3,100	−1,500	−12,800	−3,000	−24,000	−4,800	−34,800	−6,600	−57,500
TG	46,800	42,000	37,200	33,300	19,600	17,500	−9,200	−10,500	−18,000	−20,400	−25,700	−29,300	−40,800	−46,800
TH	54,000	34,000	43,000	27,500	22,000	15,000	−8,500	−11,500	−16,500	−22,300	−24,500	−31,800	−38,100	−49,700
TJ	60,000	30,000	48,000	24,000	25,000	13,000	−7,000	−13,000	−14,000	−24,500	−21,000	−35,300	−33,500	−54,600
TK	73,500	18,000	58,500	14,500	30,000	8,000	−4,500	−16,000	−8,800	−30,000	−12,800	−43,200	−21,200	−66,400
UG	77,400	72,600	61,900	58,000	33,100	31,000	−16,000	−17,200	−30,600	−33,000	−44,500	−48,100	−67,800	−73,800
UH	79,800	70,200	63,900	56,100	35,200	29,400	−15,400	−18,000	−29,400	−34,500	−42,700	−49,900	−64,800	−76,800
UJ	88,000	52,500	70,500	42,000	36,800	22,900	−12,800	−19,000	−25,000	−36,000	−37,700	−52,200	−61,400	−80,400
UK	110,000	40,500	87,000	33,000	45,000	18,000	−10,000	−22,500	−19,900	−42,000	−30,000	−59,800	−50,000	−91,800

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - OS
Air Force - 85
DSA - ES

User activities:
Army -
Navy - AS, SH, MC
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the procedure for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined.

10.2 Qualification categories. Qualification shall be categorized as follows:

Category I - Qualification shall be in accordance with the requirements of this specification.

1/ Category II - Evaluation of data, similar to that obtained from qualification testing, shall be used by the qualifying activity to establish listing. Category II listing shall not obviate the quality conformance requirements of this specification.

20. SUBMISSION

20.1 Sample.

20.1.1 Single-style submission. A sample of the size required in table V, of the highest capacitance value in each voltage rating in each operating temperature range and voltage-temperature limit in each style for which qualification is sought shall be submitted.

20.1.2 Combined style submission. Samples as shown in table X in each qualification group for which qualification is sought shall be submitted.

20.2 Test data. When inspections are to be performed at a Government laboratory, prior to submission, all sample units shall be subjected to all of the examinations and tests indicated as nondestructive in table V. Each submission shall be accompanied by the test data obtained from these inspections. The performance of the destructive tests by the manufacturer on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 Certification of material. When submitting samples for qualification, the manufacturer shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

20.4 Description of items. The manufacturer shall submit a detailed description of the capacitors being submitted for inspection, including body, coating, electrode material, terminal leads, etc.

30. EXTENT OF QUALIFICATION

30.1 Single-style submission. Capacitance-range qualification will be restricted to values equal to and less than the capacitance value submitted. Capacitance-tolerance qualification will be restricted to tolerances equal to and wider than the tolerance submitted. DC rated voltage qualification will be restricted to that submitted. Operating temperature range and voltage temperature limit qualification will be restricted to that submitted.

30.2 Combined style submission. Combined style qualification shall be restricted as shown in table X.

1/ Applicable to limited coordination specification sheets only.

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TABLE X. Combined style submission.

Qualification group	Type designation	Number of samples	Type designation	Number of samples	Will qualify 1/
I	CCR05CG331GM CCR05CG332GM CCR08CG183GM CCR08CG683GM	28 28 28 28	CC05CG331GM CC05CG332GM CC08CG183GM CC08CG683GM	22 22 22 22	CCR05, 06, 07, 08, 09 CC05, 06, 07, 08, 09 in CG, CH characteristic; F, G, J, K cap. tolerance; 200, 100, and 50 volts; FRL M.
II	CCR15CG331GM CCR15CG332GM CCR18CG183GM CCR18CG683GM	28 28 28 28	CC15CG331GM CC15CG332GM CC18CG183GM CC18CG683GM	22 22 22 22	CCR15, 16, 17, 18, 19 CC15, 16, 17, 18, 19 in CG, CH characteristic; F, G, J, K cap. tolerance; 200, 100, and 50 volts; FRL M.
III	CCR13CG100GM CCR14CG220GM	28 28	CC13CG100GM CC14CG220GM	22 22	CCR13, 14 CC13, 14 in CG characteristic; B,C,D,F,G,J,K cap. tolerance; 100 volts; FRL M.
IV	CCR54CG150GM CCR57CG271GM	28 28	CC54CG150GM CC57CG271GM	22 22	CCR54, 55, 56, 57 CC54, 55, 56, 57 in CG characteristic; F, G, J, K cap. tolerance; 500 volts; FRL M.
V	CCR75CG680GM CCR79CG823GM	28 28	CC75CG680GM CC79CG823GM	22 22	CCR75, 76, 77, 78, 79 CC75, 76, 77, 78, 79 in CG, CH characteristic; C, D, F, G, J, K cap. tolerance; 200, 100, and 50 volts; FRL M.
VI	CCR81CG680GM CCR83CG562GM	28 28	CC81CG680GM CC83CG562GM	22 22	CCR81, 82, 83 CC81, 82, 83 in CG, CH characteristic; G, J, K cap. tolerance; 200, 100, and 50 volts; FRL M.
VII	CCR90CG201GM CCR90CG512GM	28 28			CCR90 in CG characteristic; F, G, J, K, M cap. tolerance; 50 volts.
VIII			CC95ACG470K CC95AUJ101K CC96DCG4R7K CC97GCG4R7K CC96CCG100K CC97FCG100K CC96BCG180K CC96ECG180K CC96BUJ470K CC97EUJ470K	22 22 22 22 22 22 22 22 22 22	CC95 in CG, UJ characteristic; K, M cap. tolerance; 7500, 5000 volts. CC96, CC97 in CG, UJ characteristic; K tolerance; 5000 volts.

1/ ER style qualification will grant equivalent non-ER style qualification. Non-ER qualification will grant non-ER style coverage only.

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